

Intro to Neural Networks & Mechanistic Interpretability

Maths, Physics, and AI Alignment

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1. **Machine Learning**
 - 1.1 Concepts
 - 1.2 Architectures
2. **Mechanistic Interpretability**
3. **Features**
4. **Circuits**
5. **Open Problems**

Machine Learning

1. **Machine Learning**

1.1 **Concepts** ←

1.2 Architectures

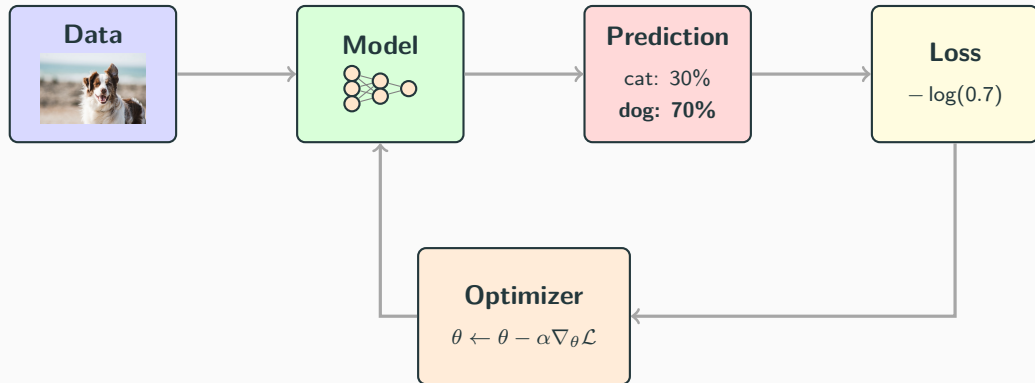
2. Mechanistic Interpretability

3. Features

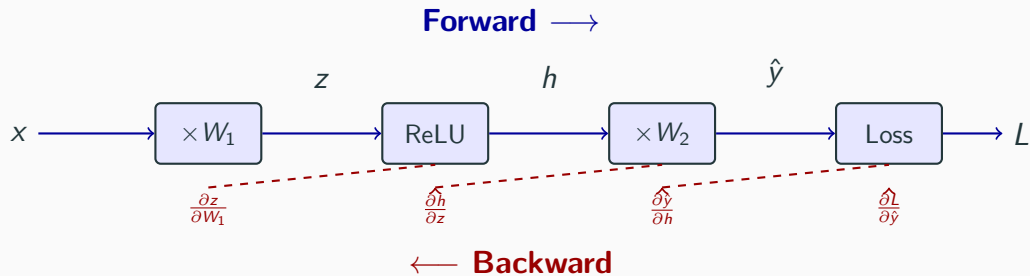
4. Circuits

5. Open Problems

The Training Loop



Forward Pass / Backward Pass



Chain rule:

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial W_1}$$

Train Loss vs. Test Loss

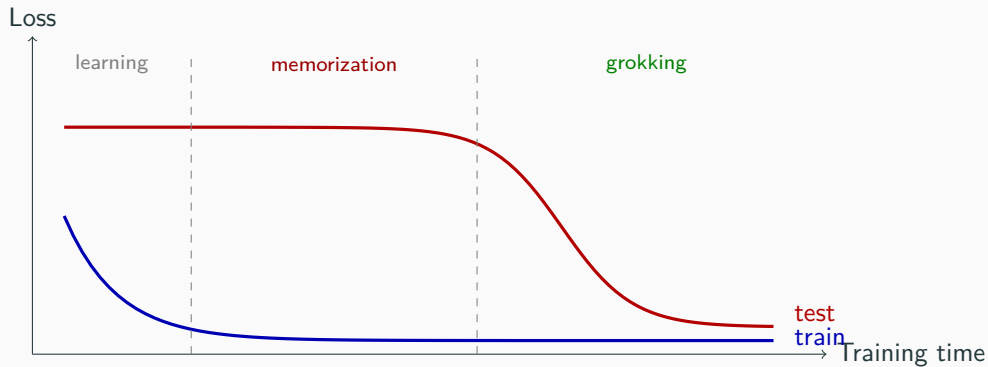


Double Descent



“Deep Double Descent: Where Bigger Models and More Data Can Hurt” [Nakkiran et al., 2019]

Grokking



"Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets" [Power et al., 2022]

Parameter / Weight the numbers that define the model, updated during training (θ , W , b)

Activation intermediate values computed during a forward pass

Hyperparameter choices made *before* training: learning rate, architecture, batch size

1. **Machine Learning**

1.1 Concepts

1.2 **Architectures** ←

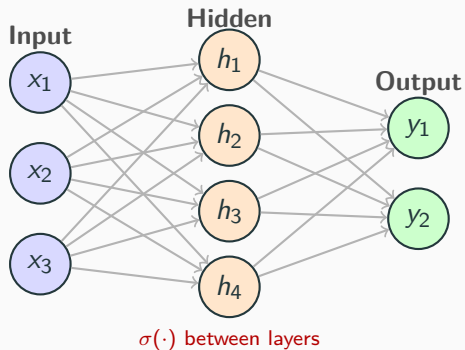
2. Mechanistic Interpretability

3. Features

4. Circuits

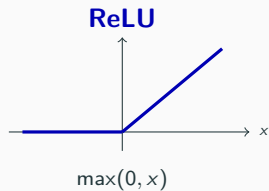
5. Open Problems

Multi-Layer Perceptron



$$h = \sigma(W_1x + b_1)$$

$$y = W_2h + b_2$$

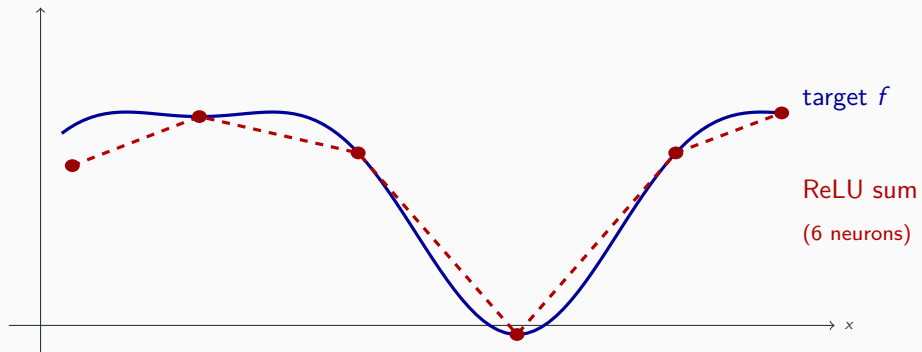


Universal Approximation Theorem

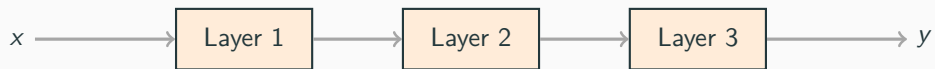
A neural network with a single hidden layer and a nonlinear activation function can approximate any continuous function on a compact set to arbitrary precision, given enough neurons.

"Approximation by Superpositions of a Sigmoidal Function" [Cybenko, 1989]

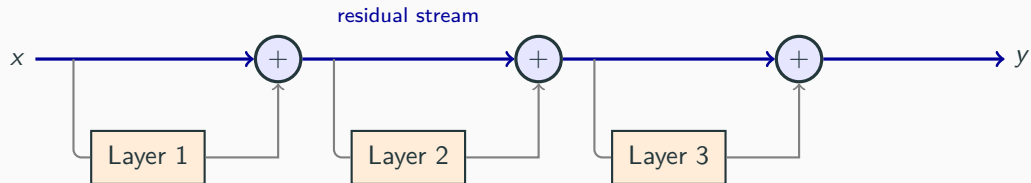
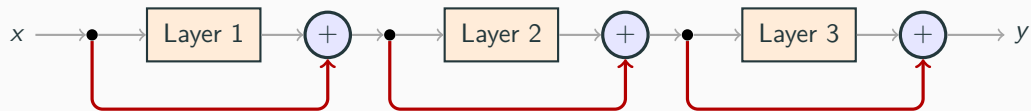
Universal Approximation: Intuition



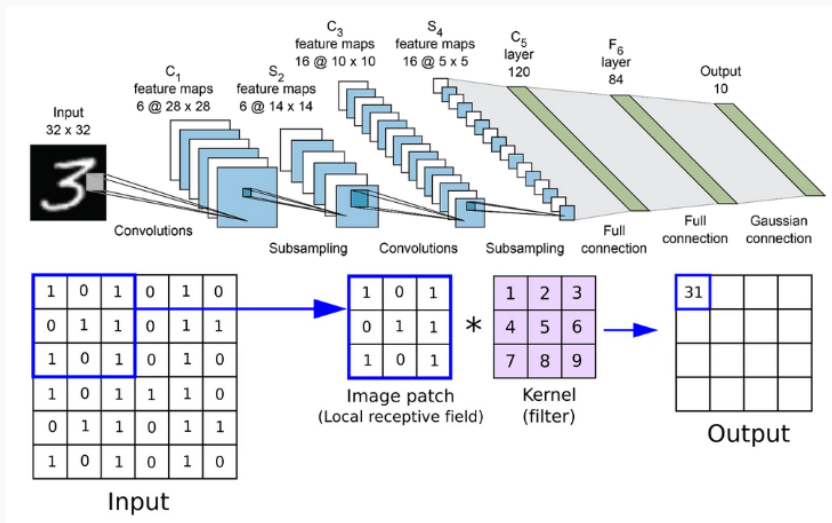
Layers



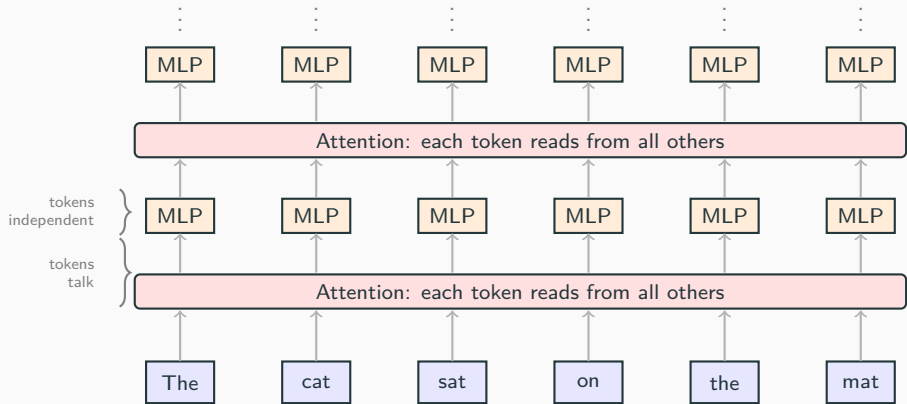
Residual Connections



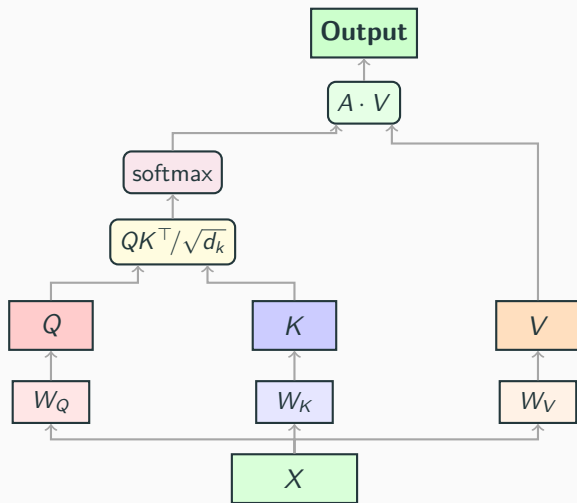
Convolutional Neural Networks



Transformers



Attention



$\text{Attention}(Q, K, V)$

$$= \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

References i

-  Cybenko, George [1989]. “Approximation by Superpositions of a Sigmoidal Function”. In: *Mathematics of Control, Signals and Systems* 2.4.
-  Nakkiran, Preetum, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever [2019]. “Deep Double Descent: Where Bigger Models and More Data Can Hurt”. In: *arXiv preprint arXiv:1912.02292*.
-  Power, Alethea, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra [2022]. “Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets”. In: *arXiv preprint arXiv:2201.02177*.
-  SuperAnnotate [2023]. *Guide to Convolutional Neural Networks*. URL: <https://www.superannotate.com/blog/guide-to-convolutional-neural-networks>.